

Low-Cost Sensing Systems for Robotics Applications

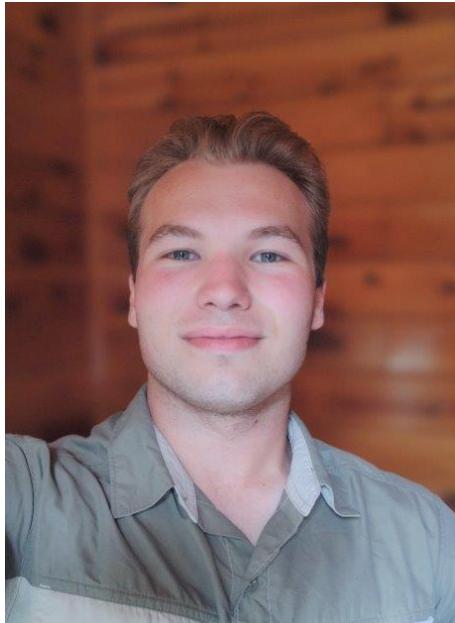
SCOPE Summit 2026



Team



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Warehouse Vs Dynamic Robotics



Warehouse

- Indoors
- Known paths (mostly)
- Predictable entities
- Well established

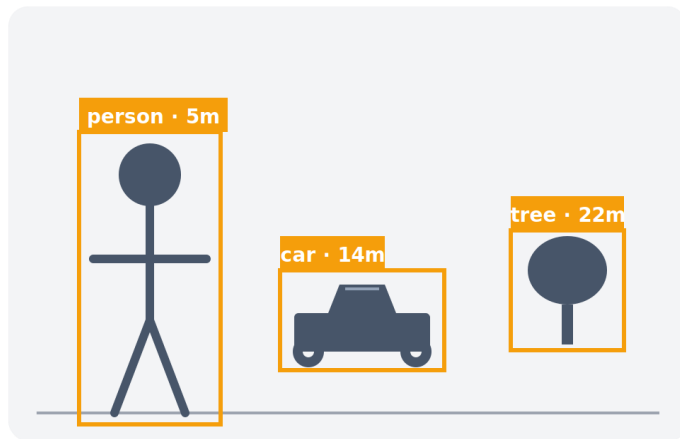


Dynamic

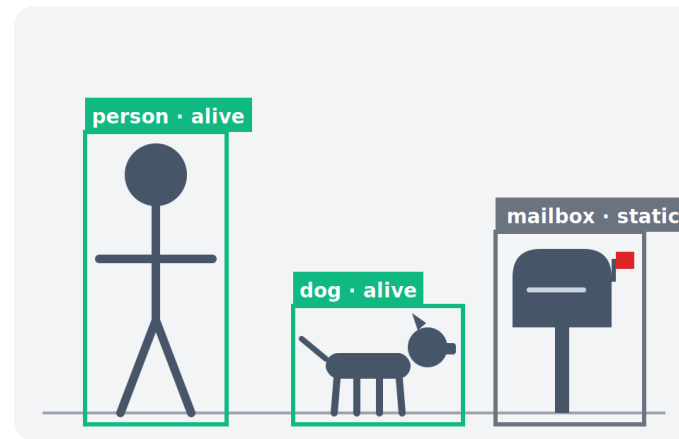
- Unknown environments
- Unpredictable entities
- Weather
- Still in development → \$\$\$\$

What do robots need to sense the world?

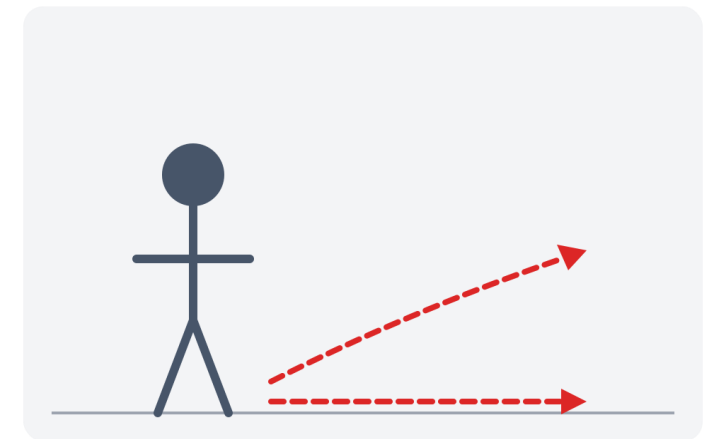
Detection



Classification



Prediction



How do robots sense the world?



Starship

Delivery robots use GPS and IMU to track location relative to others



Zipline

Cameras to target landing pads and compare that data to its GPS location to move



Waymo

Multiple LiDARs, cameras, and radars to sense the distance of its surroundings

The Goal

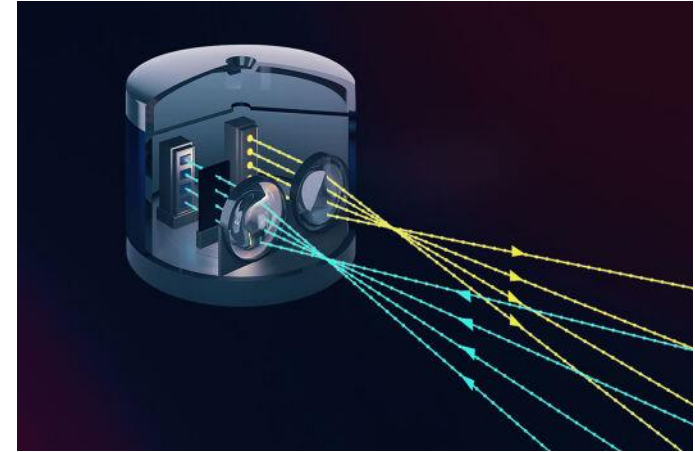
Create a series of low-cost hardware and software suites to detect, classify, and predict the motion of animate objects in a dynamic environment

What Type of Sensors We Chose



Cameras

- Low cost (relatively)
- Low fidelity depth
- Simple interfaces
- Faster integration



LiDAR

- High precision depth
- Higher cost

Hardware Configurations



HexaView Lite



HexaView Pro



HexaView Pro + LiDAR



HexaView Lite + RealSense

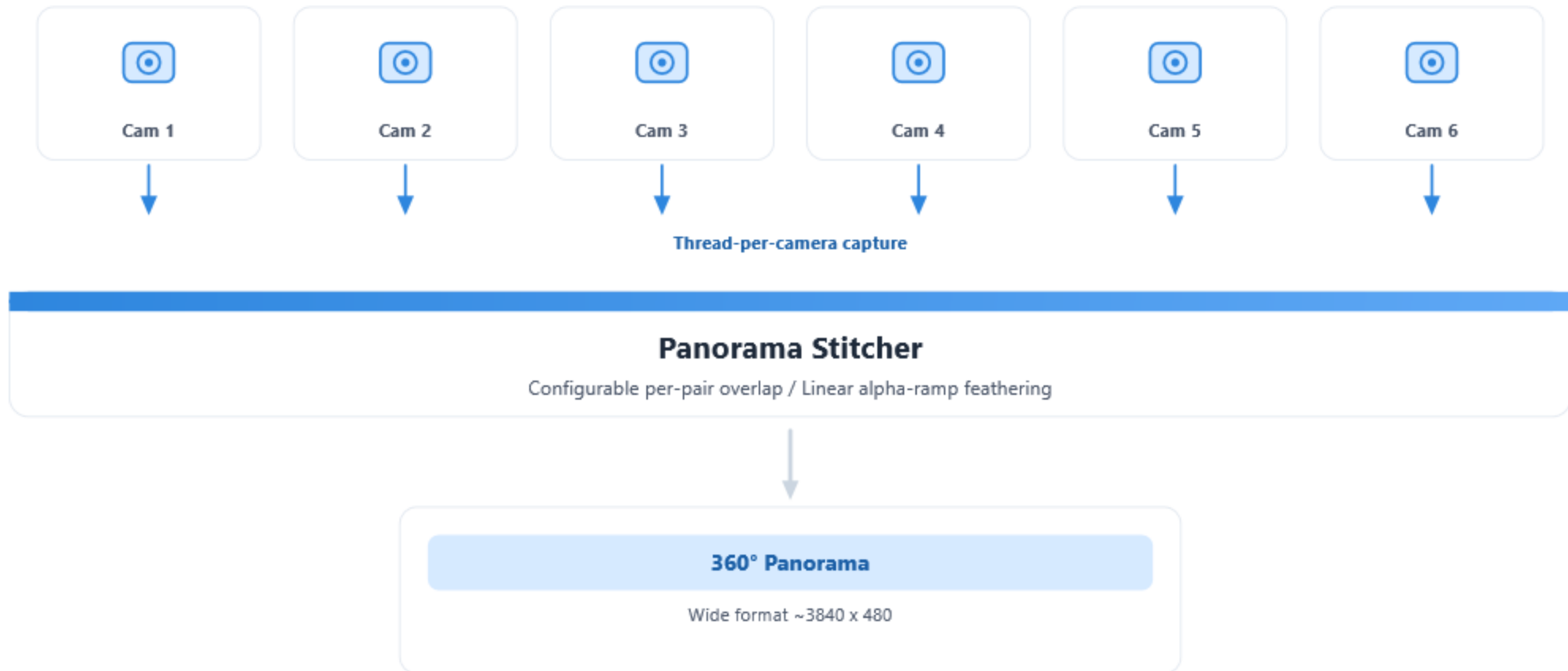


HexaRealView

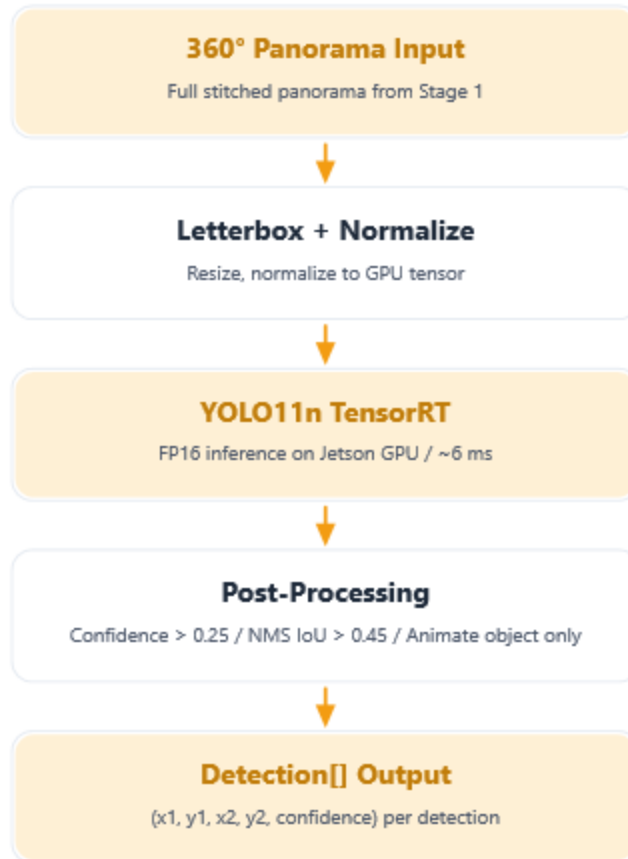


HexaRealView + LiDAR

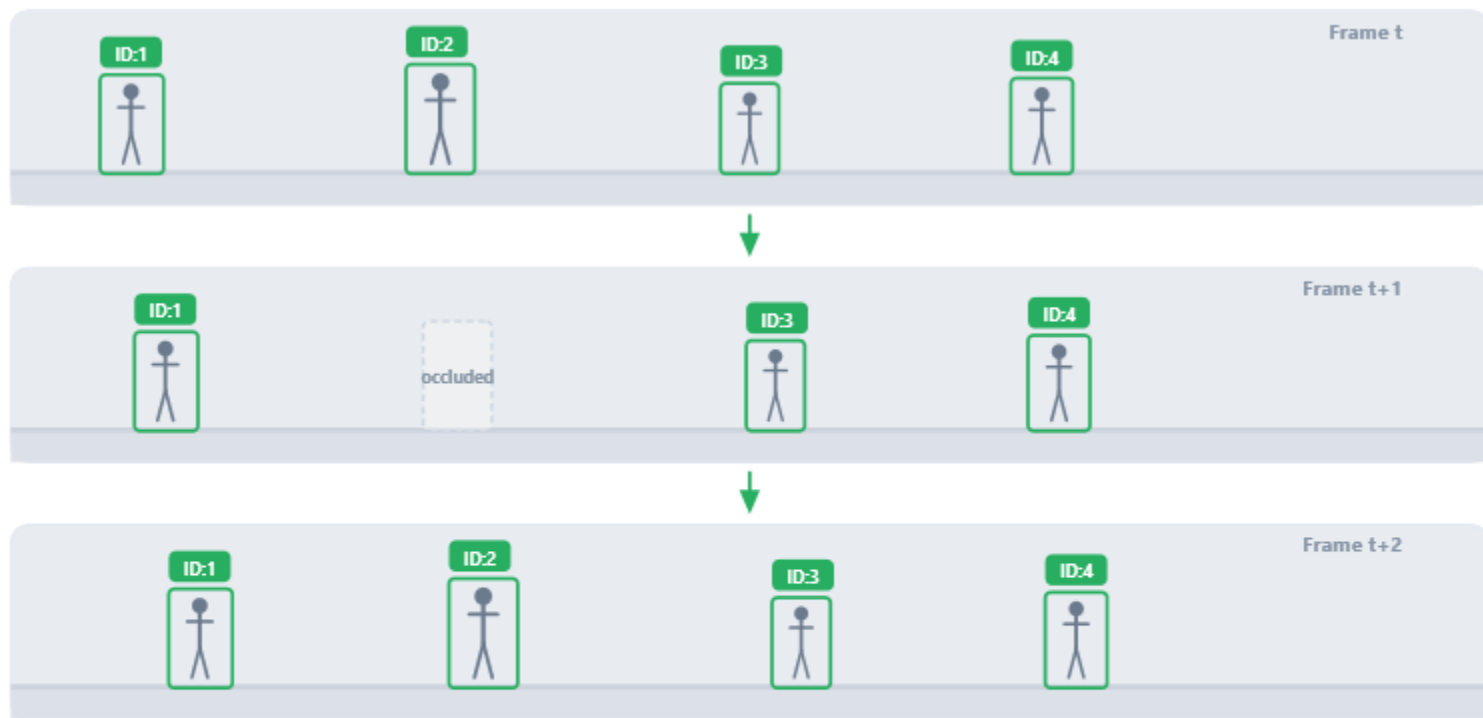
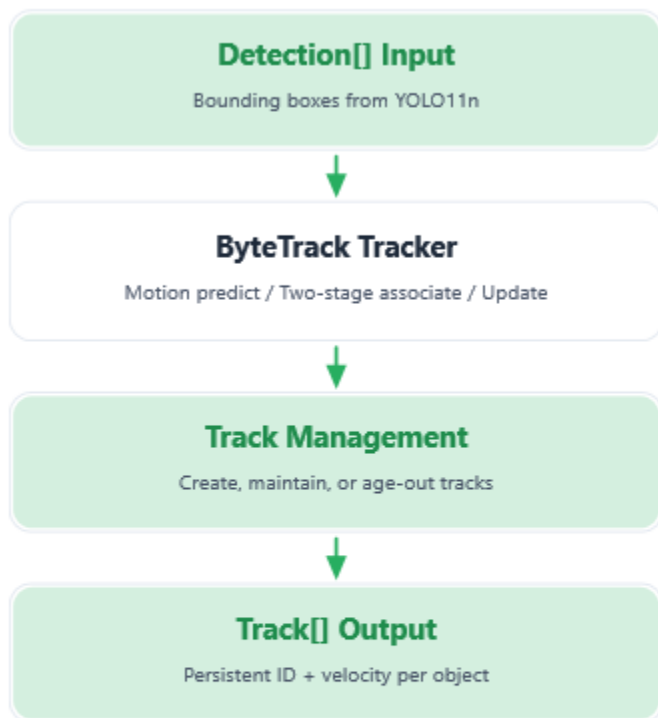
Panoramic Stitching



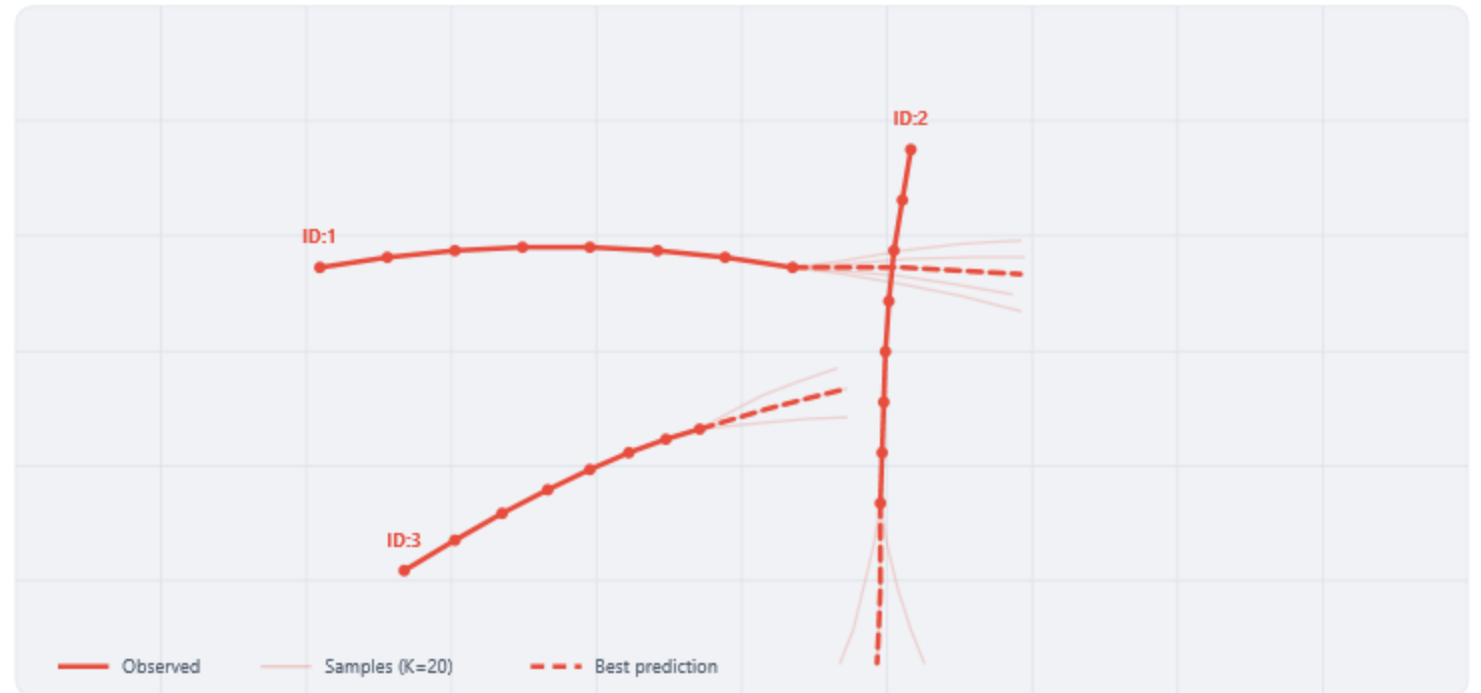
Detection



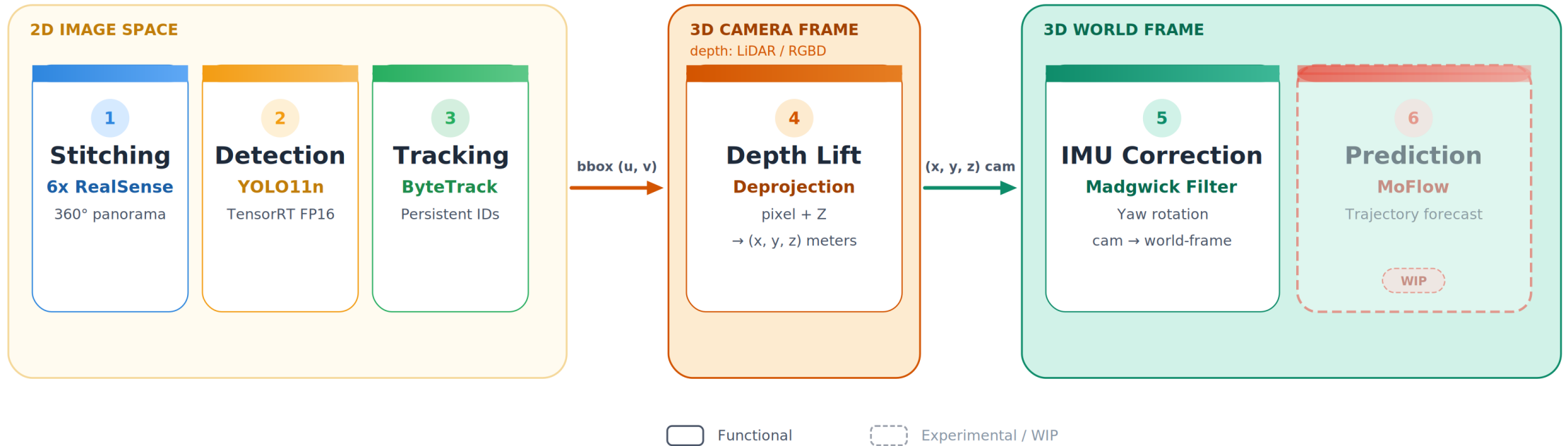
Tracking



Prediction (Experimental)



Overall Algorithmic Architecture

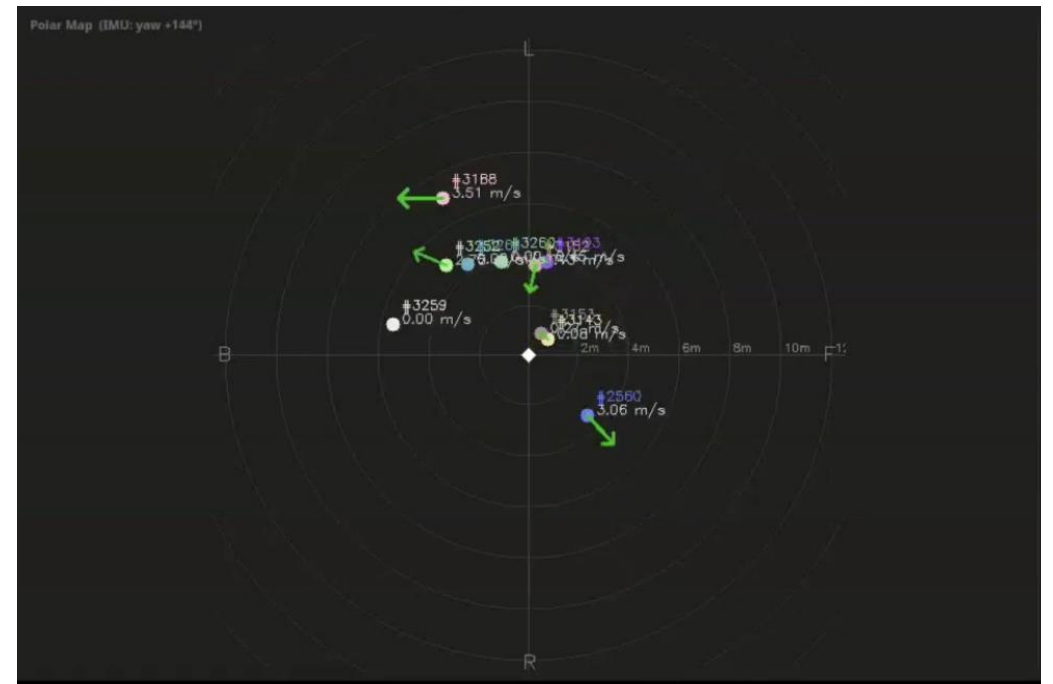
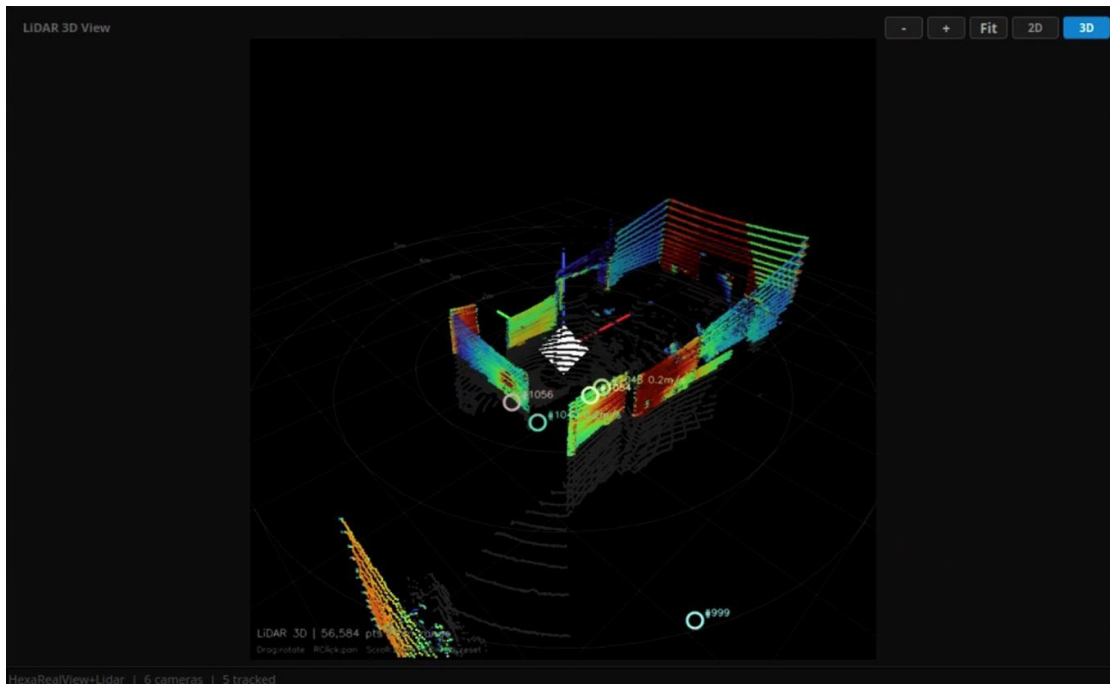


Testing Infrastructure

- Develop a modular testing system
- Individually rapidly prototype and test sensors, sensor configuration, and algorithms



Real-World Testing



Broad Configuration Tradeoffs



HexaView Lite

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- ✓ Extremely low cost
- ✗ Low frame rate
- ✗ Low quality footage



HexaView Pro

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- ✓ Better footage quality
- ✓ Still relatively low cost
- ✗ No depth sensing



HexaView Lite + RealSense

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- ✓ RGB + front-facing depth
- ✗ Footage quality mismatch
- ✗ Depth in one direction only



HexaView Pro + LiDAR

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- ✓ Precision 360° depth
- ✓ Richer tracking data
- ✗ LiDAR drives up cost



HexaRealView

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- ✓ High quality vision + depth
- ✓ 360° depth without LiDAR
- ✗ Highest cost cameras



HexaRealView + LiDAR

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- ✓ High quality vision
- ✓ Precision LiDAR depth
- ✗ Highest cost overall

Lower cost



Higher cost

Conclusion

- Off-the-shelf sensors deliver strong results at a low cost
- Proven configs start at \$360; cost scales with performance needs
- Higher-end suites require more integration but significantly boost reliability



Future Work

- Evaluate higher-quality monocular cameras
- Refine single-camera depth calibration
- Add thermal sensing capabilities
- Test & develop custom prediction models



Thank You!

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